

12.421 / 12.621

Lecture 1

Fall 2020

Material Properties

Topics covered:

- Constitutive parameters
- Permittivity and material polarization
- Simple models for permittivity of water and water ice

- Inferences from RS data

- It is often the case that we want to infer physical properties of observed areas on Earth or other bodies from RS data
- In this course (and in general) we have defined RS methods as those that utilize EM radiation
- EM radiation of interest in this course is either emitted from or has interacted with the medium of interest
- Our goal throughout most of this course then is to develop a foundational understanding of and intuition for the interaction of EM radiation with matter
- This will be done in several broad topics, starting with
 1. Constitutive parameters
 2. Classical EM principles (Maxwell's Eqs)
 3. Wave propagation

In that order

- Today, we cover constitutive parameters

- Constitution parameters

EM characteristics of any material can be described by 4 constitution parameters

★ 1. $\epsilon' \epsilon_0 \equiv$ electrical permittivity

- measure of material's ability to store an electric charge

- units F/m , where $F \equiv \text{Farad} = \frac{J}{V^2}$ is unit of capacitance

- Written above as the product of two parameters

- $\epsilon_0 = 8.85 \times 10^{-12} F/m \equiv$ permittivity of free space (minimum permittivity)

- $\epsilon' = \frac{\epsilon_{\text{material}}}{\epsilon_0}$ (dimensionless) relative (to free space) permittivity

- always the case that $\epsilon' \geq 1$

- ϵ' is of fundamental importance in RS and something we delve into more in this lecture

2. $\rho_v \equiv$ charge density

- units of $\frac{C}{m^3}$ where $C \equiv \text{Coulomb} = A \cdot s$

- $\rho_v = 0$ for RS applications $\Rightarrow$ not considered further in this course

Constitutive parameters (cont.)

3. $\mu \equiv$ magnetic permeability

- measure of material's ability to support development of magnetic field
- units $\frac{H}{m}$ where $H \equiv$ Henry $= \frac{J}{A^2}$ is a unit of inductance
- $\mu = \mu_0$ in virtually all RS applications, will not be considered further in this course

($\mu_0 \equiv$ permeability of free space $= 4\pi \times 10^{-7} \frac{H}{m}$)
 ($c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \equiv$ speed of light in a vacuum)

4. $\sigma \equiv$ conductivity

- measure of material's ability to allow current to flow through it
- accounts for EM wave attenuation in materials
 $\Rightarrow$ of fundamental importance in RS
- units of $\frac{S}{m}$ where $S \equiv$ Siemens $= \Omega^{-1} = \frac{A}{V}$
- Categories

$\sigma = 0$	"lossless"	(e.g. vacuum)
$\sigma \ll 0$	"low-loss"	(e.g. air)
$\sigma > \phi$	"lossy"	(e.g. seawater)
$\sigma \gg 1$	"conductor"	(e.g., copper, gold)
$\sigma \rightarrow \infty$	"perfect conductor"	

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• Constitutive parameters (cont.)

- For RS applications

$$\mu \approx \mu_0 \quad \text{and} \quad \rho_v = 0$$

$\Rightarrow$ mainly concerned w/ $\epsilon \epsilon_0$ and σ

- Combine these into a single, complex-valued parameter $\rightarrow$ (complex) dielectric constant (dimensionless)

$$\epsilon = \epsilon' - j\epsilon'' \quad \text{where} \quad \epsilon'' = \frac{\sigma}{\omega \epsilon_0} \quad (j^2 = -1)$$

$\omega \equiv$ (angular) frequency of EM wave

(derivation of ϵ to be done later in course)

- comments on ϵ

• ϵ' already discussed (relative permittivity)

• $\epsilon'' \equiv$ loss factor $\rightarrow$ represents attenuation of EM wave

$\hookrightarrow$ In general, RS applications involve narrow ranges of ω (and ϵ_0 is universal const.)

$\hookrightarrow$ Therefore, attenuation is governed primarily by conductivity σ .

$\hookrightarrow$ Attenuation occurs through heating $\rightarrow$ how microwave ovens work

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• Const. params (cont.)

- Comments on ϵ (cont.)

- Taking a broader view of RS, where range of ω is large, we have $\epsilon'(\omega)$ and $\sigma(\omega)$
- Several mechanisms by which ϵ' and σ vary w/ ω will be covered throughout the course, but these can be classified mainly as stretching and rotating of atoms and molecules
 - ↳ going through these will give some physical insight into the surprisingly complicated transparency of atmosphere at different EM frequencies
- For now, let's keep things simple and build some intuition

• Electric fields in matter

- Conduction is something that every one should have some broad understanding of: It is simply the ability of a material to allow the flow of electric current
- Electric current $\rightarrow$ simply the movement of charged particles; electrons and/or ions

• E Fields in matter (cont.)

- Conduction (cont)

- So, conduction is a description of the ability of electrons and/or ions to move in a material
- Metals, of course, are examples of excellent conductors
- In RS applications, seawater is the ubiquitous example of an excellent conductor
 - ⇒ we might expect the loss factor $\text{Imag}(\epsilon)$ to be high in seawater
 - more on this in a bit

- Permittivity

- In dielectric materials (i.e., insulators) charges are attached to specific atoms or molecules and are not free to "roam"
 - Charges can only move within the atom or molecule
 - Cumulative effect of these small motions account for characteristic behavior of dielectrics → permittivity
 - Remember that these motions are simply rotation and extension ...

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• E Fields in matter (cont)

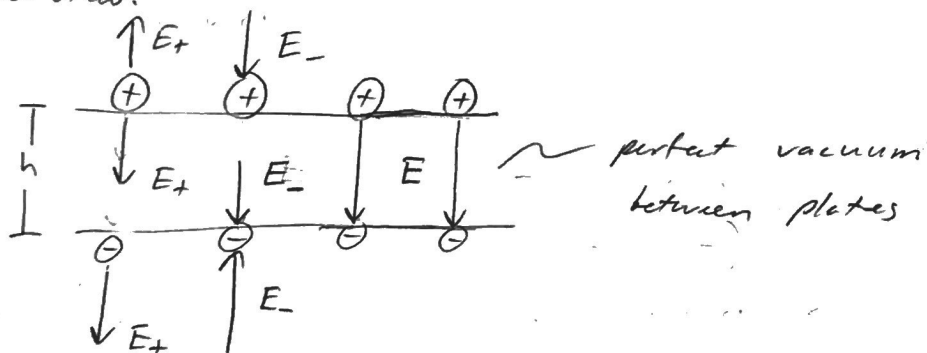
- Permittivity (cont.)

• Let's dig into this a bit more by considering the simplest case of a parallel-plate capacitor:

- Suppose we have two conductive plates of area A separated by a distance h + perfect vacuum

- Let's give the top plate a positive charge $+Q$ and the bottom plate a negative charge $-Q$

Side view:



• The $+$ charged plate produces an electric field $E_+ = \frac{Q}{2A\epsilon_0}$ that points away from the upper plate

• The $-$ charged plate does the opposite

• Thus, there is a net field $E = \frac{Q}{A\epsilon_0}$ between the two plates ($E=0$ outside the plates)

• The potential between the plates is then

$$V = \frac{Q}{A\epsilon_0} h$$

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• E fields in matter (cont)

- Permittivity (cont.)

• Parallel-plate capacitor (cont.)

- The capacitance is simply

$$C = \frac{Q}{V} = \frac{A\epsilon_0}{h} \quad (1)$$

- We can see from Eq 1 that $C \propto \epsilon_0$

- Recall that we placed a perfect vacuum between our plates

$\Rightarrow$ we might suppose that the general relation is $C \propto \epsilon' \epsilon_0$

- Recalling that $\epsilon' \geq 1$ we see that increasing permittivity increases capacitance

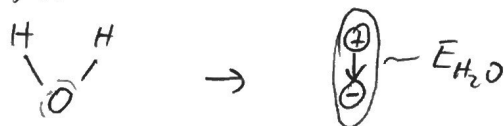
$\rightarrow$ countless observations back this up

$\rightarrow$ you may have personal experience with this as commercial capacitors always come w/ some insulator between the conductors

- To see why this is, let's fill our parallel-plate capacitor w/ a dielectric

$\rightarrow$ Let's choose a simple, ubiquitous dipole molecule: H_2O

$\rightarrow$ Because H_2O is a symmetric w/ 2 H atom oriented at an angle of 104.5° from one another



• E fields in matter (cont.)

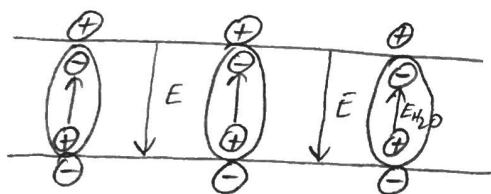
- Permittivity (cont.)

• Parallel-plate capacitor w/ dielectric

- In the absence of an external E field,

H_2O molecules are randomly oriented

- If we place them between the plates of our capacitor, orient themselves w/ the E field of capacitor



- E_{H_2O} opposes $E \Rightarrow$ reduces the net field
 $\Rightarrow$ electrical energy is stored in the "polarization" (orientation of molecules) of the dielectric

This is permittivity in a nutshell

• This is a highly simplified example to show the fundamental concepts. There are several important elements missing.

1. Heat: vibrations associated w/ heat will make it such that perfect alignment of atoms or molecules is not possible
 $\Rightarrow$ permittivity will, in general, be dependent on temperature

- permittivity (cont)

2. Impurities: Rarely in RS will you look at pure materials. Impurities or other inclusions have major influences on the effective permittivity.

E.g. Electrolytic solutions like salt water where ionic salts increase conductivity by providing charges (ions) that can move large distances $\Rightarrow$ electric current

E.g. Heterogeneous mixtures like wet or dry soil and snow

3. Dynamics: Previous example was static, but in RS care about response of materials to oscillating electric fields

- Because the field is oscillating, need to consider the response time of atoms and molecules to E field
- Will call this response time "relaxation time"

- Different mechanisms that influence permittivity will have different relaxation times $\Rightarrow \omega$. dominant mechanism governed by ω

- Electronic or atomic polarization: displacement of electron orbits relative to nucleus $\rightarrow$ highest frequencies

- Ionic polarization: displacement of ion bonds in crystals $\rightarrow$ High frequencies

- permittivity (cont.)

...

- dipolar (just discussed in capacitor example):
orientation of molecular dipoles $\rightarrow$ low freq.

- interface charge: alignment of dipole field
along grain boundaries $\rightarrow$ lowest freq

break

begin
video

pt. 2

• Simple model for response of materials to
oscillating fields, approximating using classical mechanics

- To build some intuition, let us consider that
electrons in an atom, molecules in a crystal,
dipoles in a mixture are bound by a
force that can be treated (for small
displacements) as a spring w/ constant K_{spring}

$$F_{\text{bond}} = -K_{\text{spring}} x = -m\omega_0^2 x$$

where m is the mass of our particle and
 ω_0^2 the natural frequency

- Let us also suppose there will be some
damping mechanism (friction-like force as
in our capacitance example above, radiation
in the case of atomic excitations, etc.;
all will be covered in due course
during the term)

Damping must act opposite the velocity of
the particle and we assume (will justify
later in course) it is proportional to velocity

$$F_{\text{damping}} = -m\gamma \frac{dx}{dt}$$

- permittivity (cont)

- Driving force is related to our oscillating electric field

$$F_{\text{driving}} = qE(t) = qEe^{j\omega t}$$

where q is the charge

- Newton's 2nd law

$$m \frac{d^2 x}{dt^2} = F_{\text{bond}} + F_{\text{damping}} + F_{\text{driving}}$$

$$\Rightarrow \frac{q}{m} E e^{j\omega t} = \frac{d^2 x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x$$

damped harmonic oscillator

- This eq. will have a solution $x(t) = x e^{j\omega t}$

$$\Rightarrow \frac{q}{m} E = -\omega^2 x + j\omega \gamma x + \omega_0^2 x$$

- Solving for x yields

$$x = \frac{\frac{q}{m} E}{\omega_0^2 - \omega^2 + j\omega \gamma} \quad (2)$$

- Let us define a dipole moment

$$p = qx$$

And the polarization per unit volume

$$P = Np = Nqx \quad N \equiv \text{number of dipoles per volume}$$

In all cases encountered in RS, we have

$$P = \epsilon_0 (\epsilon - 1) E$$

$\Rightarrow$ From 2, we can solve for the dielectric

constant

$$\epsilon(\omega) = 1 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 + j\omega \gamma}$$

where $\omega_p = \frac{Nq^2}{\epsilon_0 m}$ is a constant frequency

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- permittivity (cont.)

- Recalling our earlier definition of dielectric const

$$\epsilon(\omega) = \epsilon'(\omega) - j\epsilon''(\omega)$$

we can write the permittivity ϵ' and loss factor ϵ''

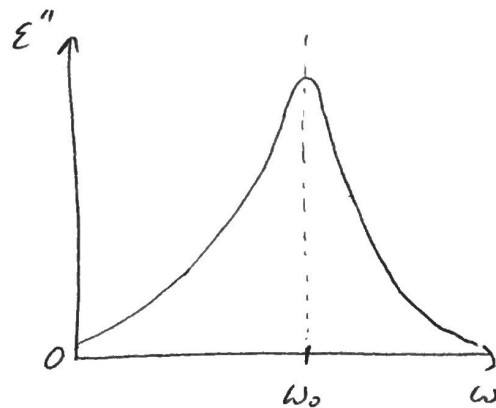
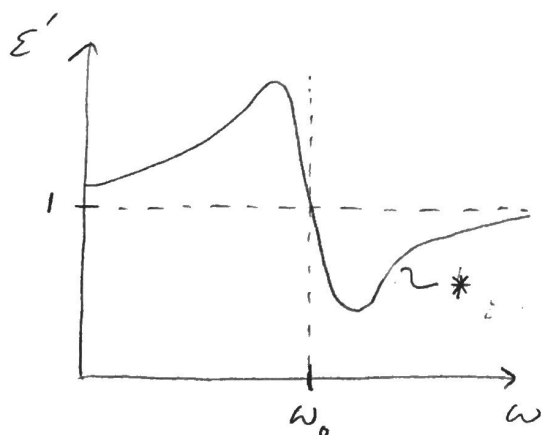
$$\epsilon' = 1 + \frac{\omega_p^2(\omega_0^2 - \omega^2)}{(\omega^2 - \omega_0^2)^2 + (\omega\gamma)^2}$$

$$\epsilon'' = \frac{\omega_p^2 \omega \gamma}{(\omega^2 - \omega_0^2)^2 + (\omega\gamma)^2}$$

Two points jump out of these equations

$$1. \epsilon' = 1 \text{ when } \omega = \omega_0$$

$$2. \text{Maximum value of } \epsilon'' \text{ is when } \omega = \omega_0$$

* Note that $\epsilon' < 1$ just above resonance freq, eventhough we mentioned before that $\epsilon' \geq 1$.

Is this a problem? Not really. First, our model is highly simplified, other mechanisms can act to increase ϵ' in real materials. Second, we'll cover later, but suffice to say for now that energy travels at a max speed, and the above does not interfere with that.

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• Properties of Natural Materials

- Preceding discussion is a simplified model meant to build intuition. Like any classical approach to quantum phenomena, it is an approximation. But turns out to be a really good one. More rigorous approaches using quantum mechanics yield similar results for the form of $\epsilon(\omega)$
- It's also worth pointing out that you can take essentially the same approach and then apply Ohm's Law to get an expression for conductivity $\sigma(\omega)$
- Dependence on physical properties like temperature T is implicit in the previous model. This will come into the damping coefficient γ and natural frequency ω_0 .
 - For T -dependence: We discussed before how atomic vibrations (heat) influence the polarizability (alignment of charges along electric field) of materials
 - ↳ permittivity is a description of polarizability, thus will be a function of T

• Prop of nat. mat. (cont.)

- T -dependence of conductivity varies depending on the material, primarily whether heat is conducted by carrier charges or through mechanistic processes (e.g., phonons)

We won't get into details in this course but suffice to say that conductivity will have some T -dependence. Classic example is for metals, where

$$\sigma = \frac{K}{LT} \quad (\text{cf. Drude Theory})$$

$K \equiv$ thermal conductivity

$$L = \frac{3}{2} \left(\frac{K_B}{e} \right)^2 \equiv \text{Lorenz number} \approx 1.11 \times 10^{-8} \frac{\text{W}\Omega}{\text{K}^2}$$

$K_B \equiv$ Boltzmann const

$e \equiv$ electron charge

This relation only holds for metals, but again, we should expect that conductivity depends on T for all materials

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- Practical example: H_2O : pure liquid, liquid w/salt, and solid

In RS applications, it is common to focus on specific freq ranges corresponding to the instrument and to consider dielectric properties in terms of a characteristic relaxation time $\tilde{\tau} \propto \frac{1}{\gamma}$ corresponding to a specific polarization mechanism:

- Pure (distilled) water at microwave frequencies ($\frac{\omega}{2\pi} \leq 50 \text{ GHz}$)

- Most common model used in RS (at microwave freq.) is the Debye model (named for Peter Debye). This takes a form very similar to the model we derived for harmonic oscillators

$$\epsilon = \epsilon_{\infty} + \frac{\epsilon_0 - \epsilon_{\infty}}{1 + j\omega\tilde{\tau}_w} = \epsilon' - j\epsilon''$$

where

$$\epsilon_{\infty} = \epsilon(\omega \rightarrow \infty) \equiv \text{high-freq limit}$$

$$\epsilon_0 = \epsilon(\omega = 0) \equiv \text{nominal or low-freq limit } (\epsilon_0 \neq \epsilon_0)$$

$$\tilde{\tau}_w \equiv \text{relaxation time}$$

- We can break into real and imag components

$$\epsilon' = \epsilon_{\infty} + \frac{\epsilon_0 - \epsilon_{\infty}}{1 + (\omega\tilde{\tau}_w)^2} \quad \epsilon'' = \frac{\omega\tilde{\tau}_w(\epsilon_0 - \epsilon_{\infty})}{1 + (\omega\tilde{\tau}_w)^2}$$

(note the similarity with harmonic model)

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- Pure water (cont.)

- T -dependence comes into the model through ϵ_{w0} and τ_w , which are invariant over given range of frequencies ($\frac{\omega}{2\pi} \leq 50$ GHz)

$\epsilon_{w0}(T)$ and $\tau_w(T)$ are empirical, commonly given as 3rd-order polynomials in T
more on this in pset 1

- $\epsilon_{w\infty}$ is weakly dependent on T ; experiments show $\epsilon_{w\infty} \approx 4.9$ for all T

- See plot and further notes in slide 4 of the lecture slides

- Salt water

- To account for the effects of ionizing salt on dielectric const of water, it is common to use a so-called double-Debye dielectric model
- This model takes into account multiple polarizing mechanism and conductivity allowed by the presence of salt

$$\epsilon' = \epsilon_{\text{abs}} + \frac{\epsilon_{w0} - \epsilon_{w1}}{1 + (\omega\tau_{w1})^2} + \frac{\epsilon_{w0} - \epsilon_{w\infty}}{1 + (\omega\tau_{w2})^2}$$

$$\epsilon'' = \frac{\omega\tau_{w1}(\epsilon_{w0} - \epsilon_{w1})}{1 + (\omega\tau_{w1})^2} + \frac{\omega\tau_{w2}(\epsilon_{w0} - \epsilon_{w\infty})}{1 + (\omega\tau_{w2})^2} + \frac{\sigma}{\omega\epsilon_0}$$

(reminder $\epsilon_{w0} \neq \epsilon_0$)

↖ permittivity of free space

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- Salt water (cont.)

- In this model

$\epsilon_{\omega} = \epsilon$ at some intermediate freq (details not important here)

And all variables are given as empirical functions of T and salinity S

- See Ulaby and Long (2015) chapter 4 for details

See slide 5
in lecture
slides

- Take-away from this model is the simple summation of the effects of multiple mechanisms and the explicit treatment of conductivity in the loss - factor

→ Such models are ubiquitous in RS, so many of you will see these things again in your research

- Freshwater ice

- Let's recall that we can classify the mechanisms that govern permittivity as either rotating or stretching

↳ in our capacitor example from before, we illustrated rotation of dipoles and just discussed ^{practical, empirical} models that consider the characteristic time it takes for dipoles to rotate (relaxation time)

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- Ice (cont.)

↳ it's easy to imagine how dipoles in gas and liquid states can rotate

↳ but solids are distinguished from liquids and gases by having an orderly structure

↳ Water ice has a simple crystalline structure that gives rise to fascinating mechanical properties

→ Due to H-bonding, ice (under terrestrial conditions) forms hexagonal crystals

→ Bonds that form hex. crystals prevent rotation of H_2O molecules in response to applied electric field

- Models of permittivity in ice, therefore, essentially omit dipole rotation mechanisms. For microwave RS applications (most common for studying ice properties) this effectively means that

$\epsilon' \approx 3.2$ - virtually independent of T and ω

$$\epsilon'' = \frac{\epsilon_{i0} - \epsilon_{i\infty}}{\omega \tau_i}$$

where variables follow same basic definitions as for pure water

- Impurities can be added to this model. See slide 6 in this lecture for more details

Summary

- We just covered a ton of material
 - For some of you, this is review with a few new pieces thrown in
 - For some of you, this is almost all new
 - ↳ If you felt lost, ask questions
 - Overview of key take-aways
 - In RS, we are most interested in permittivity and conductivity of materials
 - Conductivity is measure of a material's ability to allow electric current (flow of charges, electrons or ions)
 - Permittivity is measure of capacitance, for all intents and purposes $\rightarrow$ i.e., polarizability of atoms and molecules that make up the substance
 - ↳ Multiple polarization mechanisms, all of which involve rotating or stretching and are dominant at different EM frequencies
- stretching { Highest $\omega \rightarrow$ electronic (or atomic) polarization
 High $\omega \rightarrow$ ionic polarization
- rotation { Low $\omega \rightarrow$ dipole rotation
 Lowest $\omega \rightarrow$ interface dipole rotation